Sodium phosphate buffer

Online calculator: https://www.novoprolabs.com/tools/buffer-preparations-and-recipes/sodium-phosphate-buffer

Reference: https://www.laboratorynotes.com/preparation-of-0-1m-sodium-phosphate-buffer-of-desired-ph/

0.1 M (pH 7.8)

Chemical	M.W. (g/mol)	Addition
NaH_2PO_4	119.98	0.985 g
Na_2HPO_4	141.96	13.029 g
ddH_2O		1 L

1. Prepare 0.8 L of distilled water in a suitable container.
2. Add **0.985 g** of Sodium dihydrogen phosphate (**NaH_2PO_4**) to the solution.
3. Add **13.029 g** of Sodium phosphate dibasic (**Na_2HPO_4**) to the solution.
4. Add distilled water until the volume is 1 L.
5. Phosphate buffer is highly water soluble and has a high buffering capacity, but will inhibit enzymatic activity and precipitates in ethanol.

or prepare the stock solution separately,

1 M Sodium dihydrogen phosphate (NaH_2PO_4) → 119.98 g / L

1 M Sodium phosphate dibasic (Na_2HPO_4) → 141.96 g / L

Stock solution	Addition
1 M NaH_2PO_4 (119.98 g / L)	10.4 mL
1 M Na_2HPO_4 (141.96 g / L)	89.6 mL
Total	100 mL

0.1 M (pH 7.4)

Chemical	M.W. (g/mol)	Addition
NaH_2PO_4	119.98	0.985 g
Na_2HPO_4	141.96	13.029 g
ddH_2O		1 L

1. Prepare 0.8 L of distilled water in a suitable container.

2. Add **2.231 g** of Sodium dihydrogen phosphate (**NaH_2PO_4**) to the solution.
3. Add **11.555 g** of Sodium phosphate dibasic (**Na_2HPO_4**) to the solution.
4. Add distilled water until the volume is 1 L.
5. Phosphate buffer is highly water soluble and has a high buffering capacity, but will inhibit enzymatic activity and precipitates in ethanol.

or prepare the stock solution separately,

1 M Sodium dihydrogen phosphate (NaH_2PO_4) → 119.98 g / L
1 M Sodium phosphate dibasic (Na_2HPO_4) → 141.96 g / L

Stock solution	**Addition**
1 M NaH_2PO_4 (119.98 g / L)	10.4 mL
1 M Na_2HPO_4 (141.96 g / L)	89.6 mL
Total	100 mL

0.1 M (pH 7.0)

Chemical	**M.W. (g/mol)**	**Addition**
NaH_2PO_4	119.98	4.47 g
Na_2HPO_4	141.96	8.906 g
ddH_2O		1 L

1. Prepare 0.8 L of distilled water in a suitable container.
2. Add **4.47 g** of sodium dihydrogen phosphate (**NaH_2PO_4**) to the solution.
3. Add **8.906 g** of sodium phosphate dibasic (**Na_2HPO_4**) to the solution.
4. Add distilled water until the volume is 1 L.
5. Phosphate buffer is highly water soluble and has a high buffering capacity, but will inhibit enzymatic activity and precipitates in ethanol.